Cardiac Deformation Lab

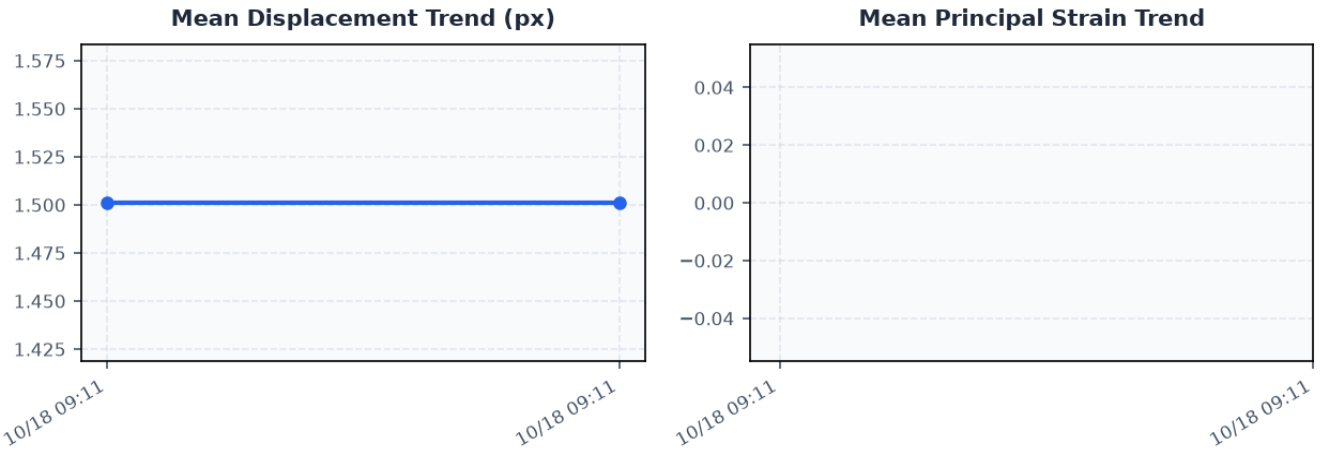
Longitudinal Multi-Session Study Analysis & Kinematic Progression Summary

Study Title:	**Report Test Study 7e94d8**	**Subject Identifier:**	TEST-REP-01
Study ID:	71964e6b-1f20-4743-9a1f-e4c7364f06d6	**Session Count:**	**2** analyses
Baseline Date:	Oct 18, 2026 09:11	**Latest Date:**	Oct 18, 2026 09:11
Protocol Notes:	Validation study for Step 11.18 reporting and export package.	**Generated At:**	2026-10-18 09:11 UTC

1. Longitudinal Analysis Session Timeline

Session / Date	Analysis ID	Type	Mean Disp (px)	Mean Strain	Principal Strain	QC Score
#1 • Oct 18, 2026 09:11	13ab0561	Single Pair	1.5011	N/A	N/A	**98/100**
#2 • Oct 18, 2026 09:11	e013a06c	Single Pair	1.5011	N/A	N/A	**98/100**

Longitudinal Kinematic Trajectories:



2. First vs. Latest Session Quantitative Delta

Deformation Metric	Baseline Value	Latest Value	Absolute Delta	Relative Change (%)	Trend Direction
Mean Displacement	1.5011 px	1.5011 px	0.0000 px	+0.0%	**Stable**
Maximum Displacement	6.6640 px	6.6640 px	0.0000 px	+0.0%	**Stable**
Mean Strain	N/A	N/A	N/A	—	**Insufficient_data**
Maximum Strain	N/A	N/A	N/A	—	**Insufficient_data**
Mean Principal Strain	N/A	N/A	N/A	—	**Insufficient_data**
Maximum Principal Strain	N/A	N/A	N/A	—	**Insufficient_data**
Anisotropy Index	0.4852	0.4852	0.0000	+0.0%	**Stable**

Strain Entropy	2.8164	2.8164	0.0000	+0.0%	**Stable**
Strain Gradient	0.0329	0.0329	0.0000	+0.0%	**Stable**
Motion Coherence	0.8749	0.8749	0.0000	+0.0%	**Stable**
Temporal Stability Index	N/A	N/A	N/A	—	**Insufficient_data**
ROI Mean Displacement	N/A	N/A	N/A	—	**Insufficient_data**
ROI Mean Strain	N/A	N/A	N/A	—	**Insufficient_data**
ROI Principal Strain	N/A	N/A	N/A	—	**Insufficient_data**
Computational Quality Score	98.0000 /100	98.0000 /100	0.0000 /100	+0.0%	**Stable**

3. Longitudinal Descriptive Statistics & Trajectory Trends

Tracked Metric	N (Obs)	Min	Max	Mean	Std Dev	Direction
Mean Displacement	2	1.5011	1.5011	1.5011	0.0000	**Stable**
Maximum Displacement	2	6.6640	6.6640	6.6640	0.0000	**Stable**
Anisotropy Index	2	0.4852	0.4852	0.4852	0.0000	**Stable**
Strain Entropy	2	2.8164	2.8164	2.8164	0.0000	**Stable**
Strain Gradient	2	0.0329	0.0329	0.0329	0.0000	**Stable**
Motion Coherence	2	0.8749	0.8749	0.8749	0.0000	**Stable**
Computational Quality Score	2	98.0000	98.0000	98.0000	0.0000	**Stable**

4. Longitudinal Quality & Protocol Consistency

Mean QC Score:	**98.0 / 100**	**Quality Tiers:**	High: 2 | Mod: 0 | Low: 0
Calibration:	**CONSISTENT**: Consistent physical calibration applied across all sessions (0.5 x 0.5 mm).	**ROI Scope:**	**NONE**: All sessions evaluated full-image deformation fields.

5. Technical Interpretation & Research Disclaimer

Methodological Scope: Longitudinal study analysis evaluates numerical trajectory variations in image-derived displacement fields, Green-Lagrange strain tensors, and kinematic coherence across multiple discrete analysis sessions. All trends represent algorithmic differences computed under classical continuum mechanics and dense Farnebäck optical flow.

RESEARCH ONLY — NO CLINICAL DIAGNOSTIC OR PROGRESSION ASSERTIONS:

Longitudinal analysis describes computational changes in image-derived measurements across analysis sessions. It is not a clinical diagnosis or clinical progression assessment. Categorizations such as 'increasing', 'decreasing', or 'stable' denote mathematical differentials and must NOT be interpreted as cardiac recovery, clinical deterioration, or medical efficacy. This software is an experimental research prototype and must NOT be used as a medical diagnostic device.